

Scenario 1:

Flight Delay Analysis

An airline tracks flight delays (in minutes) for 20 flights. Analyze the flight delays to calculate percentiles, detect outliers, and evaluate the overall distribution.

Answer:

Using np.percentile – to find 25th, 50th, 75th, 90th, quartile value to analyse how flight delays has been worked at different level.

For outlier detection, we can use IQR, lower bound and upper bound

Use a **box plot** to visualize the **spread, median, quartiles, and potential outliers** of flight delays. Use a **histogram** to assess the **distribution shape, skewness, and frequency** of delays.

Scenario 2:

Employee Salary Analysis

A company wants to analyze the salary distribution of its employees to understand the central tendency and determine whether the data is skewed.

Answer:

Central Tendency:

Mean - To find the average salary of employees.

Median - To find the middle (center) value of employee salaries.

Mode - To find the most frequently occurring salary.

Mean = median → Data is **Symmetrical / Approximately Normal Distribution**

Mean < Median → Data is **Left Skewed (Negative Skewness)**
(Few employees with very low salaries pull the mean downward.)

Mean > Median → Data is **Right Skewed (Positive Skewness)**
(Few employees with very high salaries pull the mean upward.)

Best Representation:

- Use **Median** when salary data contains **outliers or extreme values**, because it is less affected by unusually high or low salaries and gives a better measure of central tendency.

Scenario 3:

Product Sales Analysis

A retail store records product sales over 15 days. Create a frequency distribution table and visualize the sales data using appropriate charts.

Answer:

```
def freqTable(columnName, dataset):
```

```
    freqTable = pd.DataFrame(columns=["Unique_values", "Frequency",  
                                     "Relative_Frequency", "Cumsum"])
```

```
    freqTable["Unique_values"] = dataset[columnName].value_counts().index
```

```
    freqTable["Frequency"] = dataset[columnName].value_counts().values
```

```
    freqTable["Relative_Frequency"] = (freqTable["Frequency"] /  
                                       len(dataset[columnName]))
```

```
    freqTable["Cumsum"] = freqTable["Relative_Frequency"].cumsum()
```

```
    return freqTable
```

- **value_counts()** → Calculates how many times each sales value appears (Frequency Distribution)
- **Relative Frequency** → Relative frequency of observations.
- **cumsum()** → Computes cumulative relative frequency.

For visualization:

- **Histogram** → Shows the distribution of sales values and helps identify skewness and possible outliers.
 - **Bar Plot** → Displays the frequency of each sales value/category clearly.
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Scenario 4:

Student Exam Performance Analysis

A school wants to analyze the exam performance of students across three subjects: Mathematics, Science, and English. How can Data Science concepts be applied to understand their performance?

Answer:

To analyze student exam performance in **Mathematics, Science, and English**, Data Science concepts can be applied as follows:

1. Descriptive Statistics (Central Tendency)

- **Mean** → Finds the average score of students in each subject.
- **Median** → Identifies the middle score and is useful when outliers exist.
- **Mode** → Shows the most frequently occurring score.

2. Measure of Dispersion

- **Range** → Difference between highest and lowest score.
- **Variance / Standard Deviation** → Measures how spread out student scores are.

3. Data Visualization

- **Histogram** → Shows score distribution.
- **Box Plot** → Identifies outliers and spread.
- **Bar Chart** → Compares average performance across subjects.

4. Performance Comparison

- Compare subject-wise averages to identify strengths and weaker subjects.

5. Correlation Analysis

- Examine whether performance in one subject relates to another (for example, Mathematics and Science).

Scenario 5:

Clinical Trial for Diabetes Medication

A pharmaceutical company conducted a clinical trial with two groups: one receiving medication and the other a placebo. Perform a hypothesis test to determine the effectiveness of the medication.

Answer:

To determine whether the medication is effective, perform a **Hypothesis Test (Independent t-test)** because there are two independent groups:

- Group 1 → Patients receiving **medication**
- Group 2 → Patients receiving **placebo**

Step 1: Define Hypotheses

- **Null Hypothesis (H_0):** There is **no significant difference** between medication and placebo groups (medication is not effective).
- **Alternative Hypothesis (H_1):** There is a **significant difference** between the two groups (medication is effective).

Step 2: Perform Independent t-test

```
from scipy.stats import ttest_ind

t_stat, p_value = ttest_ind(medication, placebo)

print("P-value:", p_value)
```

Step 3: Decision Rule

- If **$p \leq 0.05$** → **Reject H_0** → Evidence suggests the medication has a significant effect.
 - If **$p > 0.05$** → **Fail to reject H_0** → Not enough evidence to conclude the medication is effective.
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